Name: _____ Period: _____
Earth and Space Science

Quiz 2 Study Guide: Lake Nyos Concepts

Directions: Your quiz is about a different lake, so you will need to use what you learned in the Lake Nyos lab, not memorize the lab. Study Part 1, then answer Part 2 without looking back. Check your answers with your notes.

Part 1: Key Ideas

Idea	What you need to know	Where you saw it
Latitude and longitude	Latitude is north or south of the equator (0°). Longitude is east or west of the Prime Meridian (0°). Latitude is written first, and the letter (N, S, E, W) matters.	Day 1, Task 1
Earth's systems	Geosphere (rock, magma), hydrosphere (water), atmosphere (air), biosphere (living things). Matter moves between them.	Day 3, Tasks 2–3
Density	How much mass is packed into a space. Denser material sinks below less dense material. Cold water and salty water are denser.	Demo 1, milk jug
Stratification	Water separated into layers that do not mix because each layer has a different density.	Demos 2 and 3
Solubility	How much gas a liquid can hold. Cold water holds more gas than warm water.	Demo 4 (soda)
Pressure and gas	High pressure keeps gas dissolved. When pressure drops, gas comes out as bubbles.	Soda can, Day 3 loop
CO_2 gas	CO_2 is denser than air, so it sinks and flows into low areas.	Milk jug and candles
Feedback loops	Positive: the change speeds itself up. Negative: the change slows itself down. Positive does not mean good.	Day 3, Task 5
CER	Claim: your answer. Evidence: facts or data. Reasoning: why the evidence proves the claim.	Day 4

Part 2: Connect the Ideas

1. Lake Monoun is located at 5°35′ N, 10°35′ E. Is it north or south of the equator? East or west of the Prime Meridian?

2. At Lake Nyos, cold water stayed at the bottom under warm water. Explain why, using the word *density*.

3. Two cups of water are the same temperature, but one has salt dissolved in it. Which water would sink below the other? Why?

4. A sealed bottle of soda has no bubbles until it is opened. What changed when it was opened? How is this like deep lake water that is pushed upward?

5. Fill in the Earth's systems. CO_2 from magma moving into lake water: _____ to _____.

Burning gasoline (made from oil in rock) in a car: _____ to _____.

6. Label each loop **positive** or **negative**.

(a) Rising water forms bubbles, becomes less dense, and rises even faster:

(b) Degassing pipes remove gas over many years, so the chance of a sudden release keeps dropping:

7. Why did the Lake Nyos gas cloud kill people in valleys miles away while people on higher ground survived?

8. Why can a lake near the equator store dissolved gas much longer than a lake near Buffalo, New York?

9. A student says, “The deep water is 40% saturated, so bubbles will start forming soon.” At what saturation do bubbles form on their own? Is the student correct?

How to answer a constructed response
1. Answer the question in the first sentence.
2. Use a specific piece of evidence, such as a number from a table or what a graph shows.
3. Explain why, using a science idea (density, pressure, solubility, stratification, or feedback).

Answer Key

1. North (the N) and east (the E).
2. Cold water is denser than warm water, so it sinks and stays at the bottom. The layers have different densities, so they do not mix (stratification).
3. The salty water sinks. Dissolved salt adds mass, so salty water is denser than fresh water at the same temperature.
4. Opening it lowered the pressure, so the dissolved gas came out as bubbles. Deep water pushed upward also has less pressure on it, so its gas comes out of solution.
5. Geosphere to hydrosphere. Geosphere to atmosphere.
6. (a) Positive. (b) Negative.
7. CO_2 is denser than air, so the cloud sank and flowed downhill into the valleys instead of rising and spreading out.
8. Near the equator, the surface stays warm all year, so the layers never flip and mix, and the gas stays trapped. A lake near Buffalo cools in winter and mixes twice a year, so deep water reaches the surface and the gas escapes.
9. Bubbles form on their own at 100% saturation. The student is not correct, because 40% is far below 100%.